Research Methodology in CS - Mid-Semester Examination

Answers Only

Q1. Steps involved in a research process

1. Identify and define the research problem clearly.
2. Review existing literature to understand prior work and identify gaps.
3. Formulate research questions, objectives and, where appropriate, hypotheses.
4. Design the methodology: research approach, variables, sampling, tools and procedures.
5. Collect relevant and reliable data systematically.
6. Analyze the data using suitable qualitative, quantitative or computational methods.
7. Interpret the results in relation to the objectives and existing knowledge.
8. Draw conclusions, state limitations and suggest future work.
9. Report and communicate the research with proper citations and ethical practices.

Q2. Necessity of defining a research problem

A clearly defined research problem gives the study direction and prevents vague or unfocused investigation. It specifies what is to be studied, helps establish scope and objectives, guides the choice of methodology and data, and makes it possible to judge feasibility. A precise problem also supports literature searching, hypothesis formulation and meaningful interpretation of results. Without proper definition, time and resources may be spent collecting irrelevant evidence or answering the wrong question.

Q3. Plagiarism and how it can be avoided

Plagiarism: presenting another person's words, ideas, code, figures, data or other intellectual work as one's own without appropriate acknowledgement.

Avoidance: cite every borrowed idea or result; use quotation marks and citations for exact wording; paraphrase genuinely by understanding and restructuring the idea rather than merely replacing words; cite the original source even after paraphrasing; keep accurate reference records; acknowledge reused code, figures and datasets; and check the final work for missing or inconsistent citations.

Q4. Fact-finding, analysis and evaluation

Yes. Research depends on more than collecting facts. Fact-finding supplies dependable evidence; analysis organizes and examines that evidence to reveal patterns, differences and relationships; and evaluation judges the significance, validity and implications of the findings. For example, in an algorithm study, merely recording execution times is fact-finding. Comparing the measurements under controlled workloads is analysis, while deciding whether the algorithm satisfies the research objective and discussing limitations is evaluation. Together these stages turn observations into defensible knowledge.

Q5. Literature review and an effective literature-study approach

Literature review: a systematic and critical examination of existing scholarly work related to a research problem. It summarizes current knowledge, compares methods and findings, identifies disagreements and limitations, and reveals research gaps.

Effective approach: define the research question; choose academic databases such as IEEE Xplore, ACM DL, Scopus or Google Scholar; develop keywords, synonyms and Boolean search strings; screen titles and abstracts using relevance criteria; read selected papers critically; record citations and key details; use backward/forward citation chaining; group studies by themes or methods; compare and synthesize findings; identify the gap; and maintain references consistently.

Q6. Important research designs used in experimental hypothesis-testing research

Important experimental designs include:
1. Completely Randomized Design (CRD): experimental units are assigned randomly to treatments. It is simple and suitable when units are reasonably homogeneous.
2. Randomized Block Design (RBD): units are divided into relatively homogeneous blocks and treatments are randomized within each block. Blocking reduces variation due to a known nuisance factor.
3. Latin Square Design: treatments are arranged so each treatment occurs once in every row and column, controlling two nuisance variables simultaneously.
4. Factorial Design: two or more factors are varied together. It estimates both individual factor effects and interactions,

making it efficient for studying complex systems.
5. Pre-test/Post-test Control Group Design: experimental and control groups are measured before and after treatment, helping estimate the treatment effect while accounting for initial differences.
6. Post-test-only Control Group Design: randomized groups are compared after treatment when a pre-test is unnecessary or may influence behavior.

A sound experimental design uses control, randomization and replication to reduce bias, estimate experimental error and permit valid testing of hypotheses.

Q7(a). Formulation and purpose of a research hypothesis

A research hypothesis is a clear, testable statement predicting a relationship or difference between variables. Formulation normally begins with the research problem and literature review, identifies relevant variables and the population/context, states the expected relationship, and converts it into testable form. Statistical studies commonly specify a null hypothesis (H0: no effect/relationship) and an alternative hypothesis (H1: an effect/relationship exists). A good hypothesis is specific, measurable, logically derived, feasible and falsifiable.

Its purpose is to focus the investigation, identify what evidence must be collected, guide research design and statistical testing, connect theory with observation, and provide an objective basis for accepting or rejecting a proposed explanation.

Q7(b). Deductive and inductive research

Deductive research moves from general theory or rule to a specific testable prediction. The researcher begins with theory, derives a hypothesis, gathers evidence and tests whether the prediction follows. If premises are true and reasoning is valid, a deductive conclusion is logically certain.

Inductive research moves from specific observations toward patterns and broader generalizations or theory. The conclusion is supported by evidence but remains probable rather than logically guaranteed.

Example: applying a proven algorithmic theorem to a particular graph is deductive; observing the same performance trend across many datasets and proposing a general rule is inductive.

Q8. Fundamental, Basic, Applied and Industrial Research

(a) Fundamental Research: seeks new principles, theories or knowledge without requiring an immediate practical application. Example: studying the theoretical limits of computation.

(b) Basic Research: expands understanding of underlying phenomena and concepts; the term is often used closely with fundamental research. Example: investigating why a new learning architecture generalizes under particular assumptions.

(c) Applied Research: uses knowledge to solve a specific practical problem. Example: developing an ML intrusion-detection model to reduce false alarms in enterprise networks.

(d) Industrial Research: research directed toward practical innovation, processes or products in an industrial setting, often under cost, reliability and deployment constraints. Example: evaluating a low-latency anomaly-detection system for a telecom operator's production network.

Q9. Inference on effectiveness of the anthrax vaccine

Observed table: inoculated goats - 15 survived, 2 died (17 total); non-inoculated goats - 4 survived, 10 died (14 total).

Survival rate among inoculated goats = 15/17 = **88.24%**.
Survival rate among non-inoculated goats = 4/14 = **28.57%**.
Death rate among inoculated goats = 2/17 = **11.76%**.
Death rate among non-inoculated goats = 10/14 = **71.43%**.

For a 2 x 2 chi-square test, the expected counts under no association are approximately 10.42, 6.58, 8.58 and 5.42. The Pearson chi-square statistic is about **11.48** with 1 degree of freedom ($p < 0.001$). Thus survival is strongly associated with inoculation in this sample. The inoculated group has a much higher survival proportion, providing strong evidence that the vaccine was effective under the conditions of the experiment.

Q9 OR (a). Necessity of defining a Research Problem

Defining the research problem converts a broad topic into a clear, researchable question. It fixes the scope, identifies relevant variables and evidence, guides objectives and methodology, helps assess novelty and feasibility, and prevents wasteful collection of irrelevant data. A well-defined problem also provides a standard against which the final results can be interpreted.

Q9 OR (b). Ways of conducting literature surveys in modern times

Modern literature surveys use academic digital libraries and indexing databases such as IEEE Xplore, ACM Digital Library, Scopus, Web of Science and Google Scholar. Researchers construct keyword/synonym lists and Boolean queries, apply year/subject filters, screen titles and abstracts, and maintain search records. Backward citation chaining finds foundational work, while forward citation chaining finds later developments. Search alerts keep the review current. Reference managers such as Zotero or Mendeley organize papers and citations, while literature-mapping tools can reveal themes, citation networks and research trends.

Q10(a). Experience survey

An experience survey is an exploratory method in which knowledgeable people are consulted to gain insight into a research problem. Experts, practitioners or experienced users may be interviewed using flexible questions. It helps clarify the problem, discover important variables, identify practical constraints and generate hypotheses when formal evidence is initially limited.

Q10(b). Pilot survey

A pilot survey is a small-scale trial conducted before the main survey or study. It tests the questionnaire, sampling procedure, wording, timing and data-collection process. It helps identify ambiguity, missing response options, technical problems and feasibility issues, allowing the researcher to revise the instrument before committing full resources.

Q10(c). Components of a research problem

A research problem normally contains: the problem or difficulty requiring investigation; the population, system or context in which it occurs; relevant variables or concepts and their possible relationships; the research gap or reason existing knowledge is insufficient; clear objectives/questions; and boundaries such as dataset, location, technology and timeframe. It should also be researchable, feasible, relevant and capable of producing analyzable evidence.

Q10(d). Rephrasing the research problem

Rephrasing means converting a broad, ambiguous or impractical problem into a precise research statement or question. The researcher clarifies key terms, narrows the population/context and variables, specifies measurable outcomes, removes unsupported assumptions and checks feasibility against literature and available resources. Example: instead of 'Improve cybersecurity,' use 'How can feature selection reduce false-positive rates of ML-based network intrusion detection on a defined benchmark dataset?'

Q11(a). Three copied sentences with changed words plus citation

Yes, it can still constitute plagiarism - specifically inadequate paraphrasing or mosaic/patchwriting - if the student preserves the source's sentence structure and merely substitutes a few words. A citation acknowledges the source of the idea but does not make near-verbatim copying acceptable. The student should understand the passage, express the idea independently with substantially different wording and structure, and retain the citation. Exact borrowed wording should be placed in quotation marks.

Q11(b). Paraphrasing, similarity scores and academic integrity

Good paraphrasing can reduce text similarity because the writer expresses the source's meaning using genuinely independent wording and sentence structure. However, reducing a similarity percentage is not the academic purpose of paraphrasing. Integrity is maintained only when the original meaning is represented accurately, the source is cited, and the writer does not disguise copied language. A low similarity score does not by itself prove originality.

Q11(c). AI tools used to paraphrase entire sections

Methodological concerns include loss or distortion of technical meaning, introduction of unsupported claims, inconsistent terminology, removal or misplacement of citations, and difficulty determining what intellectual work was performed by the researcher. Extensive AI rewriting may also obscure source dependence rather than produce genuine synthesis. The researcher must verify every rewritten claim against the source, preserve citations, comply with institutional or publisher rules on AI use, disclose use where required, and remain responsible for the final text.

Q11(d). Role of citation in paraphrasing

Citation identifies the original source of an idea, lets readers verify it, gives appropriate credit and distinguishes prior knowledge from the researcher's own contribution. Paraphrasing alone does **not** eliminate the need for citation: changing wording does not transfer ownership of the underlying idea. Unless the information is common knowledge, a faithful paraphrase still requires a citation.

Q11 OR (a). Interpreting an 18% similarity report

An 18% similarity score is neither automatically acceptable nor automatically plagiarism. It may be acceptable when matches consist mainly of properly quoted/cited text, references, standard terminology, unavoidable methodological phrases or small independent fragments. It is problematic when the matched material contains uncited copying, patchwriting, copied structure, reused work without disclosure, or substantial similarity to a particular source. Therefore the matched passages and their context must be examined rather than judging only the percentage.

Q11 OR (b). Mosaic plagiarism and similarity checks

Mosaic plagiarism, or patchwriting, combines phrases, sentence structures or ideas from one or more sources with minor alterations while presenting the result as original writing. It can escape simple similarity checks when words are substituted, sentence order is changed, material is drawn from many sources, or the tool's database does not contain the original source. Human review of wording, structure, citations and source dependence is therefore necessary.

Q11 OR (c). Improper citation despite paraphrasing

A paraphrase can still be academically improper if it cites the wrong source, cites a secondary source as though it were the original, omits the citation at the point where the borrowed idea appears, misrepresents what the cited author actually concluded, or uses a citation that does not support the claim. Proper citation must be accurate, relevant and traceable to the actual source of the idea.

Q11 OR (d). Role and limitations of similarity-detection tools

Tools such as Turnitin and iThenticate compare submitted text with material in their databases and highlight matching passages. They are useful for locating possible unattributed copying and helping reviewers investigate sources. Their limitations are important: similarity is not identical to plagiarism; properly quoted material and references may match; paraphrased or mosaic plagiarism may evade detection; database coverage is incomplete; and a percentage cannot determine intent or academic context. Final judgment therefore requires human examination of the matched passages and citations.